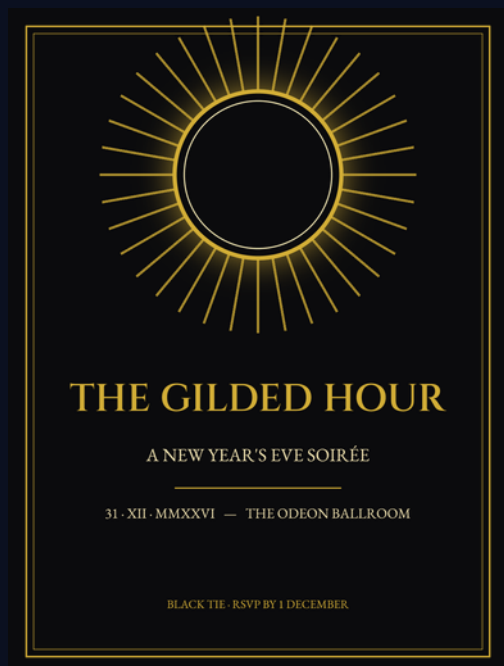


Vixl 0.22.0

Field report

The published release installed from GitHub and put through 177 recorded checks in seven areas, including full-size documents, every export format and hostile input. This report was typeset by Vixl itself.



2,832

suite tests passed

177

checks recorded

100

issues logged

0

crashes or hangs

Solid core, sharp edges

Verdict: ready for everyday work, with known sharp edges. The engine never crashed or hung in any test, and nothing unsafe got through. The project's 2,832 tests pass against the published wheel. Batches are truly atomic, documents survive 50 save/load round trips with no change, and every path-traversal, SSRF, zip-bomb and XXE probe was refused.

The defects cluster in four places:

- **Raster fills:** the PNG/raster renderer fills open stroked paths white.
- **Format parity:** SVG, PDF and PPTX diverge from the PNG output.
- **Logo package:** the workflow produces wrong output.
- **Performance on large documents:** every edit costs time in proportion to the layer count.

Most other findings are ergonomics: white default text, raw Python text in some errors, and docs that drift from behaviour.

177

checks recorded

100

issues logged

4

high severity

24

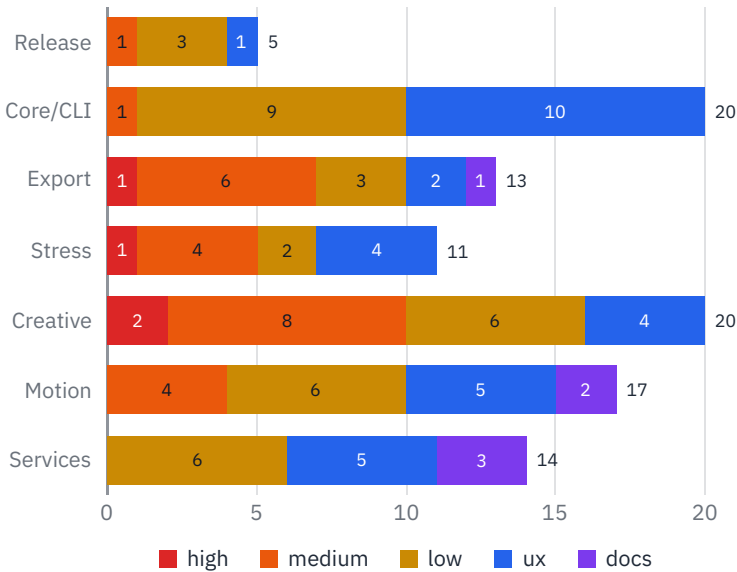
medium severity

Top issues

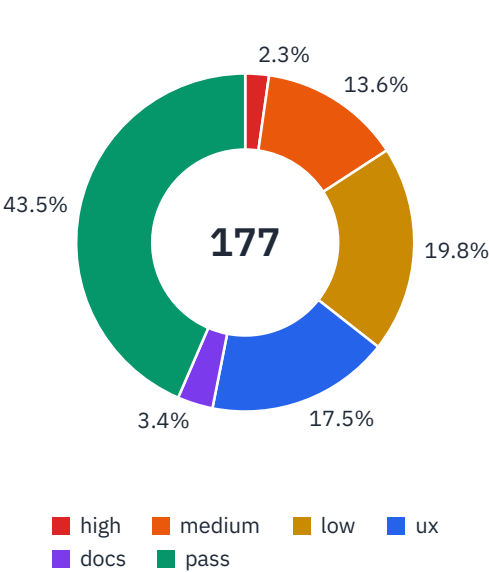
- **HIGH EXP-01** Open stroked paths are filled white in PNG/raster output whenever the path has no explicit whole-number size (the common case). SVG, PDF and PPTX are correct.
- **HIGH CRE-01/02** logo-package bakes the background into every PNG/SVG variant (mono-white is white on white), and its on-light/on-dark contrast logic is inverted.
- **HIGH STR-02** Every apply costs $O(\text{layers})$: one move on a 4,096-layer document takes about 4 s, and a 10,000-op batch on 2,000 layers takes 6 minutes.
- **MEDIUM LEAD-09 · STR-03** Some values are accepted, then make the saved document impossible to render, check or export: a fractional rich-text width, or a blur radius of 1000.
- **MEDIUM EXP-02/03/06** SVG clips strokes at the layer box. PDF and PPTX apply opacity twice on masked layers. One blend layer turns the whole SVG/PDF into a single image.
- **MEDIUM EXP-04/05** PPTX silently drops skew and the bold/italic of imported fonts.
- **MEDIUM MOT-01/04** Timeline WebP plays one loop fewer than GIF/APNG; page numbers count hidden pages.
- **MEDIUM CRE-03/05** Cyclic diagrams route edges through nodes, and the style checks flag each style's own signature moves.
- **MEDIUM STR-01** The overlap check takes about 250 s on 4,000 overlapping text labels; 4,000 shapes take 0.7 s.
- **UX CORE-06 · MOT-03** Text defaults to white, so the slides quickstart's text is invisible. Some errors leak Python internals.

Findings and measurements

Issues by area and severity



All checks by outcome



Measured numbers

Scale (stress)

- 4,096 layers: apply 4.2 s · render 11.2 s · save 0.24 s · load 1.4 s · check 24.8 s · peak 228 MB
- 4,097th layer, 10,001-op batch, 0 / negative / >16,384 px / >40 MP canvas: refused in milliseconds
- 10,000-op batch: 3.9 s on 10 layers, 364 s on 2,000 layers
- Maximum 40 MP canvas: render 28.6 s at 3.9 GB of memory
- 50 save/load round trips: 0 state and 0 pixel drift
- Concurrent writers: 120/120 CLI edits kept; Python gets write_conflict
- MP4 of 18,000 frames (600 s at 30 fps): 411 s

Interfaces (services)

- MCP tools/list core/slim: 51 tools, 58,141 chars (claimed ~58k ✓)
- all/full: 62 tools, 146,671 chars (claimed 147k ✓); compact: 13 tools ✓
- apply + check + preview 0.78 s · compose + export 1.38 s
- request_id retry 22 ms, applied once ✓
- 30+ path escapes, 10+ SSRF forms, XXE, zip-slip: all refused ✓

Fidelity (export)

- SVG vs PNG mean difference 0.1/255; PDF vs PNG 0.9/255 (no blends)
- PSD: 19 layers + 2 groups, composite equals PNG
- CMYK PDF/TIFF/JPEG: real DeviceCMYK with ink limit

Download, install, attack

Download and install

All eight assets of the **v0.22.0** GitHub release were downloaded and their checksums verified where published (only 3 of 8 assets are covered). Every archive was checked for unsafe paths. The wheel was installed with the server and pdf extras into a fresh Python 3.13 virtual environment, and every test used only that installed copy (the vixl CLI, the Python API, vixl mcp and vixl serve). The Windows installer and runtime zip were inspected but cannot run on this Linux host.

Caveats

A test suite ran alongside part of the stress testing, so absolute timings may be up to about 2x high (key timings were repeated). External HTTPS image fetches could not be tested because of the sandbox proxy. No AI providers were configured; their absence was handled cleanly.

The project's own test suite

The repository at the release commit (fb8b92f) was run against the installed wheel: **2,832 passed, 19 skipped, 0 failed** in 4 min 50 s on 4 CPUs.

Six parallel test areas

Each area was tested independently, and suspected bugs were reproduced minimally before being logged.

- **Core:** about 230 CLI calls and 15 scripts covering every layer type, text in many scripts, history, transactions, the interactive shell, fuzzed and hostile .vixl files.
- **Export:** one rich document exported to 20 format variants, verified with PIL, pypdf, pdfium, resvg, python-pptx and LibreOffice.
- **Stress:** 297 timed measurements up to the documented limits, plus concurrency and round trips.
- **Creative:** compose, rolls, layouts, 17 brushes, 33 organic presets, 11 chart kinds, diagrams, QR codes and barcodes, logo-package, proof and merge-impose.
- **Motion:** timelines, kinetic type, every animated format verified with ffmpeg, audio, forms, decks, drawing, text flow and links.
- **Services:** MCP in every toolset, --http, REST, path and SSRF probes, the updater.

Severity scale

HIGH: wrong output in a common case, or broken at a documented limit. **MEDIUM:** wrong output in a specific case, or a document left in a broken state. **LOW:** minor defect. **UX:** works, but is confusing. **DOCS:** docs disagree with behaviour. **PASS:** verified behaviour worth recording.

Release, install and test suite

Download of the published v0.22.0 assets, checksum and archive checks, install from the wheel, and the project's own test suite run against the installed wheel. *5 issues, 3 notable passes.*

LEAD-09 · rich-text with a fractional width is accepted and saved, then the document can no longer render or export

MEDIUM · Rich text / validation

Steps: python repro/richtext_float_width.py (rich-text width 300.5), then vixl -p poisoned.vixl export x.png

Expected: Fractional widths are accepted (the schema says number) or rejected at apply time.

Actual: apply and save succeed; render() and CLI export fail with 'Dimensions must be integers' (richtext.render -> Limits.size). Hit while building this report with width (W-2M)/2-20 = 510.0. Agents computing layout arithmetic will produce floats routinely.

LEAD-01 · SHA256SUMS.txt covers only 3 of 8 release assets

LOW · Release / distribution

Steps: Download v0.22.0 assets; sha256sum -c SHA256SUMS.txt

Expected: Every published asset (wheel, sdist, vixl.mcpb, vixl-claude-code.zip) has a checksum.

Actual: Only the Windows zip, the Setup.exe and vixl-update.json are listed (file uses CRLF). The wheel, sdist, .mcpb and Claude Code plugin zip cannot be verified.

LEAD-07 · add: schema allows fractional width/height but the engine rejects them

LOW · Schema / images

Steps: apply([{'type':'add', path:'img.png', width:487.3, height:302.1}])

Expected: Fractional pixels accepted (as for shapes) or the schema says integer.

Actual: VixlError 'Dimensions must be integers' from model.Limits.size; the operation schema declares width/height as number (exclusiveMinimum 0) or a percentage.

LEAD-08 · No way to write a literal \${name} in text; text-flow fails on it despite the docs

LOW · Text / variables

Steps: apply([{'type':'text-flow', name:'f', markdown:'Use \${title} in templates', frames:[{x:0,y:0,width:500,height:300}]}])

Expected: text-flow.md: '\${variables}' are not expanded while flowing, so write literal text'. A literal placeholder should be writable (an escape such as \$\$ or \${}).

Actual: VixlError missing_variable 'Undefined variable: title' raised from richtext.styled_spans during the flow's measurement. variables.substitute has no escape syntax, so documentation text that mentions placeholders cannot be typeset without a zero-width-space workaround.

LEAD-05 · vixl_guide has no 'report'/'document' kind; a report brief matches 'diagram'

UX · Agent guidance

Steps: vixl_guide(brief='a multi-page QA test report document with title page, summary stats, findings tables and image gallery, exported to PDF')

Expected: Guidance toward pages + text-flow + rich-text + charts (all of which exist and work well).

Actual: Matched kind 'diagram' and suggested big-number/bento layouts; text-flow and pages are never mentioned.

Notable passes

- **PASS LEAD-02** Release download, checksums and package structure: All listed checksums OK. No absolute or '..' entries in any zip. Version 0.22.0 in wheel METADATA, plugin.json, .mcpb manifest and .mcp.json (pinned to the v0.22.0 tag). Windows zip: 480 files, bundled CPython 3.12. sdist is 100...
- **PASS LEAD-03** Full test suite passes against the installed release wheel: 2,832 passed, 19 skipped, 0 failed in 4 min 50 s (4 CPUs).
- **PASS LEAD-06** Dogfooding: this report was produced with Vixl: Worked. text-flow reports overflow precisely (remaining_chars), so pages can be added until the story fits.

Core document model and CLI

Documents, layers, text, shapes, history, transactions, batches, the interactive shell, fuzzed input and hostile .vixl files. *20 issues, 17 notable passes.*

CORE-01 · Each apply re-measures every text layer twice; a single op on a 260-text-layer doc takes ~0.9-1.0 s

MEDIUM · performance

Steps: python perf/text_scaling.py (260 text layers in one batch: 0.8 s; then each single-op apply/move: 0.84-1.01 s). cProfile: Project.apply -> Project.inspect() called twice -> text_metrics.inspect_text x520 -> fontTools svg parse_path x3940

Expected: Per-op cost roughly independent of unrelated layers (or cached glyph metrics)

Actual: Cost is O(number of text layers) for every call; building a 130-shape labelled grid with one apply per layer took 3 min 16 s, while the same layers in one batch take <1 s. Interactive/agent loops that issue many small applies slow down linearly.

CORE-02 · Misspelled shape names are silently coerced to a different shape ('trianglee'/'hexagonn' -> 'polygon')

LOW · shapes

Steps: vixl -p P.vixl shape trianglee --name tt ; vixl -p P.vixl shape hexagonn --name hx

Expected: Error with 'did you mean triangle/hexagon?' (as 'squircl' -> squircle and 'foobarbaz' already do)

Actual: Exit 0; a 'polygon' is created and only a 'normalized' note says shape 'hexagonn' -> 'polygon'. 'trianglee' produced a full-canvas white triangle. Unrelated typos error instead, so behaviour is inconsistent. CLAUDE.md says leniency is for historical spellings, not typos.

CORE-03 · 'confetti' shape ignores the documented 'count' parameter and draws one rounded rectangle

LOW · shapes

Steps: echo

```
{ "operations": [ { "type": "shape", "shape": "confetti", "name": "c1", "x": 10, "y": 10, "width": 250, "height": 250, "fill": "orange", "count": 40 } ] } | vixl -p P.vixl apply -
```

Expected: docs/vector-paths.md lists confetti among decoration shapes taking 'count'; expected scattered pieces

Actual: A single rounded square (shape_catalog.py: confetti -> rectangle(w,h,radius)); 'count' accepted without warning. Either document confetti as a single piece or honour count.

CORE-07 · --bleed canvas is 1 px larger than physical size on each axis (rounding per side)

LOW · document

Steps: vixl new business-card --bleed -o bc2.vixl

Expected: 3.75x2.25 in at 300 dpi = 1125x675

Actual: 1126x676 (bleed 37.5 px rounded up to 38 per side). a4 --bleed --landscape --dpi 72 gives 860x613 vs 859x612.

CORE-09 · --dpi 0 silently accepted for pixel documents while --dpi -1 errors

LOW · cli

Steps: vixl new 800x600 --dpi 0 -o dpi0.vixl ; vixl new 800x600 --dpi -1 -o dpim.vixl

Expected: Both rejected with 'dpi must be 36-2400' (or both ignored)

Actual: dpi 0 exits 0 and the canvas has no dpi; -1 errors. (instagram-portrait --dpi 144 correctly errors 'pixel size')

CORE-13 · Python Project.save() silently overwrites an existing.vixl; CLI 'save' and 'new' refuse

LOW · python-api

Steps: python: Project(10,10).save('python-hello.vixl') when the file exists ; CLI: vixl -p f.vixl save other.vixl twice

Expected: Consistent overwrite policy (CLAUDE.md: refuse unless overwrite=True)

Actual: Python save replaced the file with no error; CLI save -> 'Destination already exists'. (Python export does refuse.)

CORE-14 · Python export writes RGBA PNG, CLI export of same opaque document writes RGB

LOW · python-api

Steps: python gs/py_hello.py vs vixl -p hello.vixl export hello.png

Expected: Same file format from both interfaces

Actual: Pixels identical (diff bbox None) but modes differ (RGBA vs RGB), so files differ byte-wise.

CORE-15 · polygon 'sides' bounds reported three different ways

LOW · validation

Steps: vixl shape polygon --sides 0 ; --sides 2 ; --sides 1000

Expected: One range, e.g. 3-128

Actual: 'minimum of 1' (schema), 'Polygons/stars require 3-128 sides', 'sides must be 1-128'

CORE-16 · pen --nodes help doesn't say in/out handles are absolute local coordinates; 'vixl capabilities pen' returns oil-paint first

LOW · docs

Steps: vixl pen --help ; vixl capabilities pen

Expected: Help states handle coordinate space; capabilities pen shows pen/shape/path fields

Actual: Handles are absolute (out [100,0] -> 'C103 1'); relative assumption yields loops (see showcase bezier). capabilities 'pen' maps to topic 'drawing' and lists oil-paint first.

CORE-18 · Piping output to head prints an io_error JSON ('Broken pipe') on stderr

LOW · cli

Steps: vixl -p hello.vixl inspect --json | head -30

Expected: Silent exit on SIGPIPE

Actual: {"error": "io_error", "message": "[Errno 32] Broken pipe"} (also 'ERROR: [Errno 32] Broken pipe' for vixl shapes | head)

CORE-04 · Plain 'text' has no 'width'; error suggests 'within' instead of pointing to text-layout

UX · text

Steps: apply { "type": "text", "name": "wrap", "text": "...", "width": 500 }

Expected: Hint that wrapping is set with a text-layout operation (rich-text does accept width)

Actual: "Unknown field(s) 'width' for 'text' ... Did you mean 'within' instead of 'width'?" - 'within' centres text in a shape, unrelated. rich-text accepts 'width', text does not: inconsistent.

CORE-05 · text-layout with only --width keeps a one-line height, so wrapped text is cut off

UX · text

Steps: `{ "type": "text", "name": "t", "text": "a long paragraph that should wrap inside a narrow`

`box", "size": 20 }, { "type": "text-layout", "target": "t", "width": 200 }`

Expected: Height auto-grows to fit wrapped lines when no height is given

Actual: Box becomes 200x20; result carries a warning 'does not fit its 200x20 text box ... cut off'. Warned (documented) but the obvious call produces clipped output; see `text_render.png` where 'Wrapping:' paragraph shows only line 1 and long word is truncated.

CORE-06 · Default text color is white; on a light background text is invisible unless color is given

UX · text

Steps: `vixl new 1200x1400 --background '#fbf7f0' -o T.vixl`; apply rich-text without color

Expected: Default ink chosen for contrast with background, or black

Actual: rich-text/text default to white; check reports contrast 1.07:1 (caught by check, but surprising).

CORE-08 · 'new -10x10' parsed as an option; '10X10' suggests '1080p'; " says 'Size name must be a string'

UX · cli

Steps: `vixl new -10x10 -o a.vixl`; `vixl new 10X10 -o b.vixl`; `vixl new " -o c.vixl`

Expected: 'Dimensions must be 1-16384' / accept uppercase X / 'size is empty'

Actual: 'the following arguments are required: size'; 'Unknown size 10X10; did you mean 1080p?'; 'Size name must be a string'

CORE-10 · 'undo 999' refuses entirely with 'Nothing more to undo' although 3 undo steps exist; 'undo abc' leaks Python int() message

UX · history

Steps: `vixl -p H.vixl undo 999`; `vixl -p H.vixl undo abc`

Expected: Undo as far as possible (or say 'only 3 steps available'); 'count must be an integer'

Actual: 'Nothing more to undo' (state unchanged); "invalid literal for int() with base 10: 'abc'"

CORE-11 · CLI error messages point to MCP tool names (vixl_operation_schema(...)) instead of CLI equivalents

UX · errors

Steps: `vixl -p L.vixl opacity alpha -0.1`; `vixl -p L.vixl blend alpha bogus`

Expected: 'see: vixl schema' / 'vixl capabilities blend' in CLI context

Actual: '(see vixl_operation_schema(types=['blend']))' printed by the CLI

CORE-12 · Several raw Python exception strings reach users

UX · errors

Steps: `vixl -p missing.vixl layers`; `vixl -p A.vixl apply bin.json (non-UTF8)`; `vixl -p duplayers.vixl render (layer missing 'rotation')`; `nojson.vixl (zip without project.json)`

Expected: Friendly messages: 'No such document', 'not UTF-8 JSON', 'layer x missing field rotation', 'archive has no project.json'

Actual: '[Errno 2] No such file or directory: ...'; "'utf-8' codec can't decode byte 0xff"; "Malformed Vixl archive: 'rotation'" (bare KeyError); 'Unsafe or unsupported archive member' for a zip that simply lacks project.json; duplicate *names* report 'Duplicate layer identifier'. No tracebacks though.

CORE-17 · --dry-run reports layer IDs that differ from the IDs the real apply creates

UX · apply

Steps: `vixl -p batch.vixl apply hello-ops.json --dry-run`; `vixl -p batch.vixl apply hello-ops.json`

Expected: Stable IDs or a note that dry-run IDs are provisional

Actual: dry-run `lyr_874e...`, real `lyr_c299...`; names are stable, so low impact.

CORE-19 · Unsupported export extension message doesn't list formats

UX · export

Steps: `vixl -p f.vixl export out.xyz`

Expected: 'Unsupported format .xyz; use png, jpg, webp, tiff, avif, svg, pdf, ico, pptx, html'

Actual: 'Specify a supported export format'

CORE-20 · Colour rgb(300,0,0) accepted and stored verbatim

UX · cli

Steps: `vixl -p L.vixl solid --name s2 --color 'rgb(300,0,0)'`

Expected: Clamp with a normalized note, or reject

Actual: Stored as 'rgb(300,0,0)', renders red, no note. CSS clamps, no acceptable.

Notable passes

- **PASS CORE-21** Getting-started flow from README/docs/getting-started.md runs verbatim (new, text add, align, checkpoint, inspect, check, render, export, text edit, undo, batch dry-run/apply): pass

- **PASS CORE-22** Overwrite protection: export, render --out, new, save all refuse existing files; --overwrite works; cannot export over the project: pass
- **PASS CORE-23** Batch atomicity: batch with an invalid op (missing target, unknown op, op on a layer removed earlier in the batch) changes nothing (md5 identical); all invalid ops listed together: pass
- **PASS CORE-24** Malformed JSON/NaN/Infinity/1e308/empty/non-list/huge solid in apply all rejected with clear messages, exit 1, no tracebacks: pass
- **PASS CORE-25** Corrupted .vixl: truncated, random bytes, empty, plain JSON, bad JSON, zip-bomb (2 MB -> 2 GB), path-traversal member, future format, duplicate IDs, group cycles, orphan parents, NaN opacity, unknown layer type, 100000px canvas: all rejected cleanly (<0.7 s), nothing written outside: pass
- **PASS CORE-26** Size limits: 0x0, 100000x100000, nan/inf, negative widths, dpi out of range all rejected: pass
- **PASS CORE-27** 150 named sizes as documented; letter/a4/landscape/bleed/favicon/case-insensitive names work; dpi on pixel sizes refused with a clear message: pass
- **PASS CORE-28** Layer ops: rename/duplicate (auto 'alpha copy')/hide/show/raise/lower/top/bottom/reorder/remove/group/ungroup/clip/release/mask create/invert/delete; unicode+emoji layer names; duplicate names refused: pass
- **PASS CORE-29** Opacity aliases: '70%' -> 0.7 with normalized note, 0.7 ok, bare 70 errors with an excellent hint; negative and >1 rejected: pass
- **PASS CORE-30** Typo suggestions: layer 'pannel' -> panel, blend 'multiplly' -> multiply, color 'redd' -> red, shape 'squircl' -> squircle, op 'Move'/'Target' camel/case normalized: pass
- **PASS CORE-31** History: undo/redo, checkpoints, branches, detached checkout, compare PNG, duplicate checkpoint refused, transactions begin/commit/rollback incl. nested-begin and commit-without-begin errors; a committed transaction undoes as one entry: pass
- **PASS CORE-32** Interactive shell driven via stdin: help, new, edits, errors (no abort), unterminated quote handled, EOF exits 0: pass
- **PASS CORE-33** Text shaping: Latin ligatures, Arabic joining and correct UBA ordering, Hebrew, combining marks/Zalgo; unsupported glyphs (emoji/CJK/Devanagari with bundled font) are flagged as 'fonts' errors by check rather than silently rendering tofu: pass
- **PASS CORE-34** All 130 built-in shapes render with fill/stroke; open shapes stay unfilled; arc with no angles is a full disc as documented: pass
- **PASS CORE-35** All 9 blend modes produce mathematically expected colors; radial gradient clipped to star; rich-text bold/colour/underline/strike: pass
- **PASS CORE-36** Python API parity: same 48-op batch via Python and CLI is pixel-identical; VixlError carries code, field, suggestions, operation_index; Python undo reverts a whole batch: pass

- **PASS CORE-37** --json errors are structured (error, message, field, suggestions, available, operation_index); errors go to stderr; exit code 1: pass

Export and import fidelity

One rich document exported to every format, then re-opened and compared across renderers; imports including hostile SVGs and URLs. *13 issues, 9 notable passes.*

EXP-01 · Raster renderer fills open stroked paths white when the path's size is derived (no explicit integer width/height)

HIGH · render/raster (PNG, JPEG, WebP, TIFF, PSD, previews)

Steps: vixl new 400x200 -o w.vixl --background '#3355aa'; apply {type:shape, shape:path, path:'M 0 0 Q 100 200 200 0', stroke:'#cc0000', stroke_width:6, x:100, y:50} (no width/height, or width:200 only); vixl export w.png

Expected: An open path with a stroke and no fill stays unfilled (CLAUDE.md: geometry.OPEN_SHAPES/default_fill, 'in every renderer'). PDF, SVG and PPTX leave it unfilled.

Actual: PNG/render show a white-filled region inside the curve. The same op with integer width:200,height:100 renders correctly. Root cause: design_render.shape_image sends any non-int width/height (auto-derived 200.0) to vector_strokes.render, whose primitives use layer.get('fill','white') (vector_strokes.py:415, also :320) instead of geometry.default_fill(layer). The same route also covers dashed/width-profile/marker/catalog strokes, so those open paths are probably affected too. Independently re-confirmed by the lead t...

EXP-02 · SVG export clips strokes (caps, miter joins) at the layer box

MEDIUM · export/SVG

Steps: Same doc as EXP-01 with stroke_width 20, path 'M 0 0 L 100 100 L 200 0'; export v.svg; rasterize with resvg

Expected: Stroke extends past the geometry box like it does in PNG/PDF/PPTX (red ink bbox 94,44-305,162).

Actual: Each layer is a nested <svg width/height viewBox> with no overflow='visible', so the default overflow:hidden clips it. Ink bbox is 100,50-299,149, which cuts off the end caps and the bottom miter. Vixl's own editable SVG import then rejects this file ('Clipped nested SVG strokes require svg_mode=appearance').

EXP-03 · PDF and PPTX apply layer opacity twice on masked layers

MEDIUM · export/PDF, export/PPTX

Steps: new 200x100 white; rectangle fill #000 opacity 0.5 + mask create (opaque); export m.png/m.pdf/m.svg/m.pptx; rasterize

Expected: Mid-grey (127) in every format

Actual: PNG 127 and SVG 127, but PDF 191 and PPTX (LibreOffice) 191, i.e. an effective opacity of 0.25. In the showcase doc the 'band' raster fallback's SMask max is $184 = 255 \cdot 0.85 \cdot 0.85$. Clip and layer-style fallbacks are not affected.

EXP-04 · PPTX silently drops skew

MEDIUM · export/PPTX

Steps: rectangle 60x60 + skew x:30; export pptx; inspect XML / render with LibreOffice

Expected: Skew is kept (as a freeform or picture fallback), or the layer is listed in raster_fallbacks/warnings

Actual: Written as an un-skewed prstGeom rect at the original box with no warning. Red rows are 220-279 in every row, while PNG/PDF show 205-263 to 237-295.

EXP-05 · PPTX loses bold/italic of imported font files (weight/style are not written)

MEDIUM · export/PPTX text

Steps: font import DejaVuSerif-Bold.ttf --name serifb; text with font serifb; export pptx

Expected: <a:rPr b='1'> (and i='1' for FreeSansOblique), so PowerPoint picks the bold face

Actual: Only typeface='DejaVu Serif' / 'FreeSans' is written with no b/i attributes. LibreOffice renders the title in Regular and the subtitle upright (title region differs from PNG on 34% of pixels).

EXP-06 · One blend-mode layer rasterizes the whole SVG and PDF page (text no longer selectable, no fonts)

MEDIUM · export/SVG, export/PDF

Steps: Showcase doc with one multiply and one screen layer; export doc.svg and doc.pdf

Expected: Both SVG (mix-blend-mode) and PDF (/BM Multiply, /Screen) support these blends natively; only that layer should fall back

Actual: doc.svg is a single <image> (vector_only false). doc.pdf reports raster_fallbacks {layer:'page'}, fonts:0, and has no text. Without the blends (vec.*) both are vector with 2 embedded fonts. This is reported in the metadata and docs mention 'may rasterize the document', but the fidelity cost is large and --svg-policy strict is the only guard.

EXP-08 · Very large images cannot be imported at all, even with max_pixels / --max-pixels

MEDIUM · import/image

Steps: PNG 12000x9000 (108 MP); vixl import huge.png; also apply {type:add,path:huge.png,max_pixels:4000000} with --max-pixels 200000000

Expected: Downsampled to max_pixels (the field exists for this), or a Vixl error that names the limit and the remedy

Actual: Fails with Pillow's raw message 'Image size (108000000 pixels) exceeds limit of 89478485 pixels, could be decompression bomb DOS attack.' Neither option has any effect. 108 MP is a real camera resolution.

EXP-07 · --simulate is silently ignored for PPTX export

LOW · export/PPTX

Steps: vixl -p vec.vixl export out/sim.pptx --simulate deuteranopia

Expected: Simulated colours (as PDF does: a raster page), or an error like SVG's

Actual: The file is byte-for-byte the same size as the normal export (108342 B) with identical colours and no warning

EXP-09 · Vixl's own SVG export does not re-import in editable mode

LOW · import/SVG round-trip

Steps: vixl import out/vec.svg --svg-mode editable; also a one-path export (repro/v.svg)

Expected: At least simple Vixl SVGs round-trip as editable layers

Actual: 'Referenced SVG paints require appearance/auto' (gradients) and 'Clipped nested SVG strokes' (single path). auto/appearance give one 1200x800 raster layer. Behaviour matches the docs, but the round-trip is only appearance-level.

EXP-13 · Raster TIFF has no resolution tags when the document has no dpi

LOW · export/TIFF

Steps: 1200x800 doc (no print size); export doc.tiff

Expected: 72/96 dpi tags or nothing (it is a print raster format)

Actual: No 282/283 tags (Pillow reports dpi (1,1)). Letter docs do get 300 dpi, so this matters only for custom pixel sizes.

EXP-10 · Image import error messages leak Python object reprs

UX · import/image

Steps: vixl import notanimage.jpg (text bytes); vixl import font.ttf

Expected: 'Not a supported image (JPEG/PNG/WebP/GIF/...)'

Actual: 'Cannot decode image: cannot identify image file <_io.BytesIO object at 0x7fd1dc846d90>'

EXP-12 · --cmyk --proof on PNG gives a misleading error

UX · export CLI

Steps: vixl export doc_proof.png --cmyk --proof

Expected: A hint to drop --cmyk (docs: export proof.png --proof)

Actual: 'CMYK export supports JPEG, TIFF and PDF'. With .jpg: 'Choose either a CMYK separation or an RGB soft proof' (clear). SVG --simulate also says '...for CMYK and proofing', which is a slightly wrong message.

EXP-11 · Docs say CMYK PDF is raster; it is vector

DeviceCMYK

DOCS · docs/exporting.md format table

Steps: export deck.pdf --cmyk on the pages doc

Expected: Docs match the behaviour

Actual: The output is content:vector with DeviceCMYK k/K operators, CMYK shadings and embedded fonts. exporting.md says 'CMYK PDF is raster'.

Notable passes

- **PASS EXP-P01** All formats produce valid files: PNG, JPEG, WebP, AVIF, TIFF (RGB and CMYK), SVG (well-formed XML), PDF (pypdf/pdfium), PPTX (python-pptx and LibreOffice), PSD (8BPS v1, 19 layers + 2 groups, names/opacity/blend keys correct, composite equals PNG), HTML (self-co...
- **PASS EXP-P02** SVG and PDF vector output match the PNG closely: With no blend modes: SVG via resvg mean abs diff 0.1/255, PDF via pdfium 0.9/255. Radial gaussian/smooth falloff, rotated text, ligatures and the QR/barcode match. Open paths stay unfilled in SVG/PDF/PPTX.
- **PASS EXP-P03** Overwrite protection: Every format and command (export to all formats, render, export-icons) refuses an existing target with output_exists unless --overwrite is given.
- **PASS EXP-P04** CMYK is real: The PDF uses DeviceCMYK (k operators, CMYK shading). TIFF and JPEG are mode CMYK with GCR (K>0 on 96% of pixels) and max TAC 264 under --ink-limit 280. --proof with ink-limit 150 visibly changes the colours.
- **PASS EXP-P05** Multi-page: PDF has 3 pages with TrimBox/BleedBox (9pt bleed) and embedded subset fonts; --pages 2-3 works. PPTX has 3 slides, the HTML presenter has 3 slides (svg or png), render --page all writes a labelled contact sheet, and a page-1 TIFF...
- **PASS EXP-P06** SVG import security: XXE, billion-laugh, script/onload, file:// and http:// image hrefs, relative path images, external CSS and a 1e6 viewBox are all refused before parsing or fetching. The local HTTP server logged no requests.
- **PASS EXP-P07** URL import SSRF guard: Refused with a clear message: https 127.0.0.1, localhost, ::ffff:127.0.0.1, 10.0.0.1, 169.254.169.254 and decimal 2130706433, plus http://, file:// and ftp://.
- **PASS EXP-P08** Bad image input: Truncated PNG, a non-image file and missing files give clean JSON errors with no traceback. JPEG/PNG/WebP/GIF import correctly, and a PDF page imports as a raster with a warning; an out-of-range page is refused.
- **PASS EXP-P09** vixl diff: Works on two images and on two .vixl documents, and writes a diff PNG.

Stress and performance

Layer and batch limits, huge canvases and texts, deep nesting, history growth, concurrency and save/open round trips. *11 issues, 8 notable passes.*

STR-02 · Every apply costs O(layers), so a batch costs O(ops x layers): a documented 10,000-op batch takes about 6 minutes

HIGH · apply/performance

Steps: Python API: 2,000 shape adds plus 8,000 move ops in one apply (detail='brief'). Also a single move on a 4,096-layer doc, and a scaling sweep of 100/400 moves across 10/500/2,000/4,000 layers.

Expected: A batch within the documented 10,000-op limit should finish in seconds, and one edit on a big doc should not take seconds.

Actual: 10,000 ops: 364 s (dry_run 360 s; the same batch with a bad last op ran 394 s before it was rejected). The same 10,000 ops on a 10-layer doc: 3.9 s. One move on a 4,096-layer doc: 4.0-4.5 s (repeated 4x). Sweep: about 23 ms per op at 4,000 layers vs 0.5 ms at 10 layers. In the profile, apply() runs inspect() twice (8.3 of 10.3 s), and render.resolved_layers/solve run once per op. forms.has_fields scans every layer for each text layer through document_variables (9.5M genexpr calls for one move), which makes text me...

STR-01 · Overlap check is far slower than the CHANGELOG's '<1 s at 4,000 layers' claim; it is pathological when text layers overlap

MEDIUM · checks/performance

Steps: 4,000 text layers on a 4000x4000 canvas, dense grid (labels slightly overlap neighbours): Project.check(checks=['overlap']).

Variants: 1,000 labels; 4,000 non-overlapping labels; 4,096 mixed layers (256 text).

Expected: Under 1 s at 4,000 layers (CHANGELOG 0.22, #263).

Actual: Overlapping labels: 252.8 s and 244.1 s at 4,000 layers (3.6 s / 9.4 s / 22.5 s at 250/500/1,000, so it grows superlinearly).

Non-overlapping text: 6.0 s and 6.2 s. Mixed 4,096 doc: 2.4-3.0 s.

4,000 shapes only: 0.69 s. In the profile, checks.alpha ->

render.layer_canvas_surface builds a full-canvas RGBA image (Image.new + getchannel) for every candidate text layer, which is 21 of 23.7 s at 1,000 layers. Timings were taken with a test suite running in the background, but the gap is 6x to 250x. (Lead review: downgr...

STR-03 · Blur radius up to 1000 is accepted, then the document can no longer render, check or export

MEDIUM · effects/validation

Steps: 4000x4000 canvas, 2000x2000 ellipse; apply effect blur amount=1000; then render(), export png/pdf, check().

Expected: Refuse at apply time with resource_limit (the operation is atomic and validated), or clamp the padding.

Actual: apply succeeds with no warning. render/export/check then each fail with 'Image exceeds pixel limit'. Saving works, so the broken doc is persisted, and undo is the only fix. Radii of 300/500/700 render, but peak RSS grows to 1.16 GB. There is a separate wording inconsistency: amount=100000 gives 'radius must be at most 1000' while 100001 gives 'amount must be at most 100000'.

STR-04 · text-flow of a 100k-character unbreakable word takes about 5.6 minutes; 100k lorem takes 29 s

MEDIUM · text/performance

Steps: text-flow with 4 columns in 1900x1900 at size 8, using 100,000 x 'a', then (lorem x 2000)[:100000].

Expected: Seconds; long-word breaking should not be quadratic.

Actual: One word: apply 338.7 s, then PDF export 14.4 s. Lorem: 28.6 s apply, 7.4 s PDF. 5,000 short lines: 1.2 s. Output is correct (4 frames of about 13-16k chars each).

STR-05 · A maximum-size canvas (40 MP) needs about 4 GB RSS to render; under tighter memory it fails with a raw numpy MemoryError

MEDIUM · memory

Steps: Project(16384, 2441) with gradient, blurred ellipse and text; render/export PNG/JPG; repeat under prlimit --as 2 GB and 3 GB.

Expected: A predictable footprint within typical agent/container limits, or a clean resource_limit error.

Actual: Render 28.6 s at 3.9 GB peak RSS (about 97 bytes per pixel: full-canvas float64 arrays). PNG 14.0 s, JPG 14.0 s, PDF 2.5 s, check 14.5 s. With 2 GB/3 GB caps: MemoryError 'Unable to allocate 305 MiB/1.19 GiB ... float64' (Python exception, not a VixlError; the process survives).

STR-10 · Group nesting limit is 16 levels, and a 1,000-deep chain takes 9 s to fail

LOW · limits/docs

Steps: Chain of group ops nesting one shape 14/15/16/100/500/1,000 deep in one batch.

Expected: A clean, fast refusal, with the limit documented next to the 4,096-layer limit.

Actual: Depths 14 and 15 work. Depth 16+ fails with 'Group/clipping nesting exceeds 16 levels' (no operation index), and failing takes 0.2 s / 2.3 s / 9.3 s for 100/500/1,000. The document is unchanged.

STR-11 · Several hard caps are lower than an agent might assume: repeat 512, radial-repeat 360, pen points 512, frame animation 256 frames, timeline 3,600 frames, keyframe sample duration 600 s, dpi 36-2400

LOW · limits/docs

Steps: repeat 1000/4095, radial-repeat 361/1000/100000, scatter 5001, frame-save 257th, timeline gif 120 s @30fps, keyframes duration 600001 ms.

Expected: Clean refusals.

Actual: All refuse cleanly with resource_limit/invalid_operation. 'Repeat count must be 1-512' is a different wording from radial-repeat's schema-style message. Letter at 1200 dpi passes the dpi range but fails with 'Image exceeds pixel limit' (it needs 92 MP). GIF of 120 s @30fps is refused with a clear 'in-memory pixel budget ... export mp4' hint.

STR-06 · The error for a too-long pen 'points' list echoes the whole input (1.6 MB for 100k points) and does not state the maximum (512)

UX · errors

Steps: pen operation with points lists of 513, 1,000, 2,048 and 100,000.

Expected: A short error such as 'points holds at most 512 entries; got 100000'.

Actual: Message length 8,271 / 16,063 / 32,831 / 1,600,063 chars ('[[500.0, 500.0], ...] is too long'). 512 points works. The equivalent path-d string works up to 8,192 commands with clear errors past that.

STR-07 · A text that would be taller or wider than 16384 px fails with a generic 'Dimensions must be 1-16384'

UX · errors

Steps: text op with 100,000 chars on one line, 5,000 lines at size 10, or 1,300 lines at size 10.

Expected: An error naming the text layer's computed size, suggesting text-flow/text-layout or a smaller size.

Actual: 'resource_limit Dimensions must be 1-16384' with no mention of text, the layer or the measured size.

STR-08 · undo(N)/redo(N) with N larger than the available history does nothing and says 'Nothing more to undo'

UX · history

Steps: 2,200 single edits, saved and reloaded; undo(2200), undo(1802), then undo(1801); redo(1802), then redo(1801).

Expected: Undo as many steps as exist and report the count, or say how many are available.

Actual: undo(1802) and redo(1802) fail with 'history_boundary Nothing more to undo/redo' although 1,801 steps exist. undo(1801) takes 3 ms; redo(1801) takes 18.2 s (asymmetric). The final state round-trips exactly.

STR-09 · Float text size is rejected outright (size 25.6 or 0.1) rather than normalized

UX · validation

Steps: text op with size computed as a float (25.6).

Expected: Per the normalize.py leniency convention, round it and report it under 'normalized', or at least accept integral floats.

Actual: 'size must be integer; got float 25.6'. The whole batch is rejected, with all 62 invalid ops listed (the batch-level listing is good). Font size limits 1-4096 are clean; size 1 and 4096 render.

Notable passes

- **PASS STR-12** Layer limit: 4,096 layers build, render and export; the 4,097th layer is refused atomically: 4,096 layers: apply 4.2 s, first render 11.2 s (cached re-render 1.8 s), save 0.24 s (0.44 MB), load 1.4 s, default check 24.8 s, PNG 2.3 s, SVG 0.7 s, PDF 2.2 s, peak RSS 228 MB. 1k: render 3.4 s, check 7.4 s. 2k: render 6.1 s,...
- **PASS STR-13** Batch limit: 10,001 ops refused instantly; a 10,000-op batch with an error at op 9,999 leaves the document unchanged: 10,001: refused in 0.003-0.08 s ('A batch holds at most 10000 operations, got 10001; split it'). Bad last op: rejected after 394 s with operations[9999] named; state hash unchanged. A 10,000-op batch on a small doc: 3.9 s.
- **PASS STR-14** Canvas size validation is clean: 1x1 works; 0, negative, float, >16384 and >40 MP are refused instantly; no OOM: All refusals take <1 ms with resource_limit/invalid_operation. 1x1 renders and exports. dpi outside 36-2400 is refused.
- **PASS STR-15** Concurrent editing: no corruption or lost updates (CLI file lock; Python save detects write conflicts): 8x30 distinct: 34.5 s with all 8 docs at 30 layers (1x30 sequential: 12.2 s, about 0.4 s per CLI call). 2x60 same file: 30.1 s, 120/120 layers, no errors. Python: 176 layers, all unique, every successful save present; 302 write_c...
- **PASS STR-16** 50 load/save round trips: no state or pixel drift: State drift 0/50, pixel drift 0/50, project.json identical. The zip container bytes differ on each save (zip entry timestamp only), so byte-identical saves are not guaranteed. SVG export is deterministic (5/5). 50 cycles took 8.1...
- **PASS STR-17** History: 2,200 edits keep the file small; pruned to 1,801 revisions; compact works: File size 1.5 -> 21.5 -> 41.3 -> 72.4 -> 72.3 KB (capped). Apply latency is about 16 ms average and flat. 1,801 undo steps kept (max_history 2,000 minus 10% when pruning). Undo one step at a time: 33.9 s total. compact: 3 ms, dro...
- **PASS STR-18** Long timelines export to MP4 with no frame cap; GIF caps are enforced with actionable errors: 10 s@30: GIF 6.4 s (0.8 MB, 300 frames), MP4 1.3 s. 120 s@30: GIF refused (pixel budget), MP4 28.3 s for 3,600 frames. 600 s@30: building keyframes took 77 s (slow), GIF refused (3,600-frame cap), MP4 411 s for 18,000 frames (7 M...

- **PASS STR-19** Scatter 5,000 and radial-repeat 360 work; caps are enforced: Scatter 5,000: apply 8.9 s, render 2.1 s, PDF 0.95 s, SVG 0.78 MB. radial-repeat 360: apply 0.22 s, render 3.3 s. Over-cap values are refused instantly.

Creative features

compose, roll, layouts, styles, looks, organic shapes, scatter and patterns, charts, diagrams, codes, workflows, colour tools and brushes. *20 issues, 15 notable passes.*

CRE-01 · Logo package PNG/SVG variants bake in the document background; mono-white files are white-on-white

HIGH · workflow logo-package

Steps: compose logo/fernhill.vixl (background #ffffff, mark group + wordmark), then `\`vixl workflow logo-package --request req/logopkg.json\``

Expected: Logo PNGs have alpha (transparent) and SVGs carry no background rect, so mono-white/mono-black can sit on any surface

Actual: Every PNG is mode RGB with an opaque background; mark-mono-white-512w@1x.png is 100% white pixels (extrema 255..255). SVGs begin with `<rect fill=rgb(255,255,255)>`.

CRE-02 · Variant contrast logic is inverted: on-light swaps to black, on-dark keeps a dark-green wordmark that is illegible

HIGH · workflow logo-package

Steps: Same run as CRE-01; read report.variants in logo/pkg.out; view horizontal-on-dark PNG

Expected: #163d2e wordmark on white is ~11:1, so on-light keeps full colour; on-dark should switch to the light/mono-white version

Actual: Report says 'full-colour logo average colour has only 1.2:1 contrast on #ffffff' (so on-light is replaced with black) and on-dark reports contrast 15.32, but the dark-green wordmark on #111 measures 1.53:1 (`\`vixl color contrast #163d2e #141414\``). The average colour seems to include the baked-in white background.

CRE-03 · radial/mindmap layouts draw a cyclic back-edge straight through every node; tree/grid overlap edges but check passes

MEDIUM · diagrams

Steps: diagram-from-text 'Idea -> Sketch -> Prototype -> Test / Test -> Prototype: iterate / Test -> Launch / Launch -> Measure -> Idea' with `--layout radial|mindmap|tree|grid`

Expected: Layouts route the cycle edge around nodes; check flags any edge overlapping another

Actual: radial and mindmap lay the cycle out as a straight line and Measure->Idea runs through all nodes (check --checks diagram catches this). tree and grid show coincident edges rendered as double-headed arrows (Idea<->Sketch, Launch<->Measure) yet the diagram check passes. Hub-shaped graphs render well in radial/mindmap.

CRE-04 · diagram-from-text ignores width/height; diagrams render small

MEDIUM · diagrams

Steps: In an operation batch: diagram-from-text with x=60,y=1060,width=1080,height=220 (hero/info2.vixl)

Expected: Diagram fills or fits the requested box

Actual: Layer is 764x76 at x=218; text is ~13px. In 1400x900 canvases every layout uses a small part of the canvas.

CRE-05 · Style checks flag the style's own signature moves (false positives)

MEDIUM · check/style

Steps: Compose the art-deco poster (radial-repeat sunburst, centred axis) and the brutalist poster with the style's own recommended pairing archivo-black-archivo; run `\` check --checks style\``

Expected: Art-deco sunburst and centred layout pass symmetry/upright; Archivo Black counts as a black weight

Actual: art-deco: 'symmetric 40%, wants 80%' on a mirror-symmetric poster and '32 elements rotated off the grid' (the sunburst rays the style asks for). brutalist: 'text weights [400]; wants 700+' because Archivo Black is registered at weight 400.

CRE-06 · Applying a style through compose does not make the layout or fonts fit the style; the style check then fails its own result

MEDIUM · compose/style

Steps: compose instagram-square with style=swiss + layout big-number (req/social.json), then `\` check --checks style\``

Expected: style=swiss gives flush-left text, a grotesk pairing and one accent

Actual: Text stays centred, 2 hues, bundled DejaVu fallback font. check reports swiss/text-align and swiss/hue-count warnings, plus a 'fonts: bundled fallback' warning. The style is guidance only, so the style=field looks like it does more than it does.

CRE-07 · sketch look turns flat shapes into a faint pastel wash instead of pencil lines

MEDIUM · looks

Steps: look sketch on a star, a rounded rect and text (looks/sketch.json)

Expected: Pencil-like edges or hatching

Actual: pencil-sketch effect (amount 80) leaves about 15% opacity tints with no outline. The shapes almost disappear.

CRE-09 · Auto value labels thin unevenly: the largest series is never labelled and labels drop from some bars

MEDIUM · charts

Steps: chart bar --csv charts/sales.csv (3 series) --legend bottom; then chart-data --set 'Jun:Online=260'

Expected: Either all bars are labelled (smaller font) or none, applied consistently per series

Actual: The Online series has no labels. After the data edit, Retail May/Jun lose their labels too (10 label layers instead of 18).

CRE-10 · compose cannot fill image slots from files: meme/photo layouts need a separate import step

MEDIUM · compose

Steps: compose with layout meme-top-bottom

image='memes/img0.png'

Expected: compose accepts a workspace path (or imports it) for image/images

Actual: 'ERROR: layout: Missing embedded asset: memes/img0.png'.

Only an asset id from a prior import into the same document works, so one-call meme/photo builds are impossible. Workaround: new, import, remove the import layer, then layout-apply with 'assets/<sha>.png'.

CRE-11 · Layouts' own full-bleed output fails check (safe_area errors on memes)

MEDIUM · check

Steps: Apply meme-top-bottom, meme-four-panel or meme-comparison, then `\` vixl check\``

Expected: Full-bleed meme panels are intentional and pass

Actual: check passed=false: 'panel-1 extends outside the safe area', 'top-text extends outside the safe area' etc.

CRE-08 · watercolor look is barely visible on flat vector shapes; duotone/risograph default to muddy greys

LOW · looks

Steps: looks sheet (looks/*.json)

Expected: A visible wet-edge/bleed texture; duotone and risograph use the shape's or a pleasant ink pair

Actual: watercolor is almost the same as clean-flat. duotone and risograph turn coral/teal into grey-mauve when no colour is given.

CRE-12 · Seamless pattern tiles produce 'cut off by the canvas edge' warnings for wrapped motifs

LOW · check

Steps: pattern-scatter 512x512 tile (pattern/ops.json)

Expected: Wrapped motifs in a seamless tile are marked intentional

Actual: Many bounds warnings (tile/23/m-daisy ... cut off) on the expected wrap copies

CRE-13 · compose 'background' is ignored when a layout sets its own background

LOW · compose/layout

Steps: compose poster-18x24 background #10131c + layout event-poster

Expected: The requested background stays

Actual: Rendered background is the layout palette's #203c3b; nothing in the result mentions the override

CRE-16 · Thumbnail-legibility findings are noisy on print/large pieces and chart internals

LOW · check

Steps: check on charts/all.vixl, hero/info2.vixl, deco poster

Expected: Few findings, or grouped per chart

Actual: One info/warning per chart value label ('2.1 px tall at 320 px wide'), plus a chart's own line overlapping its own value label reported as an overlap

CRE-17 · Minor rendering artifacts: hairline seams on pathfinder divide pieces; scatter 'inside' lets motifs spill past the outline

LOW · pathfinder/scatter

Steps: shapes/ops.json

Expected: Divided pieces abut without seams; copies stay inside

Actual: Thin light seam along the circle/square boundary after divide. Stars hang over the rounded-rect edge.

CRE-19 · color scale silently ignores extra colours/--count; out-of-gamut oklch clipped without notice

LOW · color

Steps: color scale '#1f6f50' '#f4efe6' --count 5 --space oklch; color convert 'oklch(70% 0.4 30)' --to hex

Expected: Two-stop 5-step scale or an error; a gamut warning

Actual: Returns a fixed 11-step 50..950 scale from the first colour only. oklch(70% 0.4 30) gives #ff6551 with no warning.

CRE-14 · Layout body/subtitle leading is very loose (~1.9x), and text spacing cannot be negative for tight display leading

UX · layouts/type

Steps: slide-title layout subtitle at 63px: baselines 121.5px apart; text op spacing -30

Expected: Leading of about 1.2-1.5 for display/subtitle; negative spacing allowed for display type

Actual: Two-line subtitles look like separate paragraphs (slide.png, poster.png body). spacing<0 is rejected: 'spacing -30 is less than the minimum of 0'.

CRE-15 · roll is reproducible but 'mood: playful' gives quiet, similar directions with small headlines

UX · roll

Steps: roll --for poster --mood playful --seed 42/7/123/2026 --apply --set title=...

Expected: Visibly different and playful directions that use the canvas

Actual: Same seed gives byte-identical renders (md5 match). The four seeds all give a small centred or right-aligned title stack on calm palettes, with about 60% of the canvas empty.

CRE-18 · ``text add`` defaults to white text; highlighter brush is opaque

UX · paint/text

Steps: text add 'round' on a cream canvas; paint --brush highlighter

Expected: Default text colour readable on the background; highlighter is translucent

Actual: Brush labels are invisible white-on-cream. The highlighter stroke is solid navy like a marker.

CRE-20 · ``vixl layers`` prints full SVG path data for every layer

UX · cli

Steps: vixl --project shapes/shapes.vixl layers

Expected: A compact listing

Actual: Several KB of path coordinates per torn/irregular layer

Notable passes

- **PASS CRE-21** compose one-call builds work (poster, social card, slide, logo, infographic); atomic errors name the step: works
- **PASS CRE-22** roll seeds are reproducible (same JSON md5 and byte-identical renders for seed 42 twice): works
- **PASS CRE-23** Font pairing works offline in this env: Lora, Source Sans 3, Literata, Cinzel, EB Garamond, Archivo Black, Fraunces and Work Sans all rendered; catalog lists 104 families. Unknown pairings suggest alternatives.: works
- **PASS CRE-24** All 33 organic presets render, are attractive and are seed-stable: works
- **PASS CRE-25** irregular (rough), tear (bottom+right with rim) and scatter preset fur look good: works
- **PASS CRE-26** Seamless tile: seam report says seamless (edge_error 3.3 vs interior 4.3); a 3x3 tiling at 2x zoom shows no visible seams: works
- **PASS CRE-27** radial-repeat mandala (7 rings, about 200 copies) renders fast and cleanly; check passes with 0 issues: works

- **PASS CRE-28** pathfinder exclude/subtract/divide give correct geometry: works
- **PASS CRE-29** All 11 chart kinds render from CSV with good defaults; chart-data set (CATEGORY:SERIES), append and remove-category work; PPTX export ok: works
- **PASS CRE-30** QR (H, version 5, Unicode URL) matches an independent segno encoding module for module (0/1369 mismatches); EAN-13 decoded from pixels to 4006381333931 (checksum ok); Code128 decoded from pixels to 'VIXL-0.22-QA 123456' (checksum ok). No zbar/cv2/pyzbar is installed, so I used custom decoders.: works
- **PASS CRE-31** merge-impose from CSV: 7 badges on A4 with crop marks, per-row QR, `role:Attendee` filled blank cells, fonts embedded: works
- **PASS CRE-32** workflow proof writes a self-contained 1.3 MB HTML with checks and decisions: works
- **PASS CRE-33** glow, neon, soft-halo, soft/hard shadow, outline, gradient, grain, paper, film, halftone, hand-made and plush all work. Text-incompatible looks give clear errors.: works
- **PASS CRE-34** All 17 brushes paint distinct, good-looking strokes: works
- **PASS CRE-35** color convert/harmony/contrast, palette-generate and type-scale work; bad input gives clear errors: works

Motion, audio, pixels, forms and decks

Timelines, kinetic type, GIF/WebP/APNG/MP4/WebM export, pixel art, audio, fillable forms, decks, drawings, text flow and linked documents.
17 issues, 13 notable passes.

MOT-01 · export-timeline WebP writes loop count N, while GIF/APNG (and export-animation WebP) encode N extra plays

MEDIUM · export/loop

Steps: repro/: 64x64 doc, timeline-set loop=2, animate translate-x; export-timeline loop2.gif/.png/.webp. Then frame-save a/b, animation-set --loop 2, export-animation frames2.gif/.png/.webp. Parse the WebP ANIM chunk (webpinfo.py) and PIL loop info.
Expected: All formats play 3 times. The pixel doc says 'loop is the number of additional repetitions ... both exports encode that same meaning'. WebP ANIM loop_count counts total plays, so it should be 3.
Actual: Timeline: GIF NETSCAPE=2, APNG num_plays=3, WebP loop_count=2 (one play short). Pixel export-animation: GIF 2, APNG 3, WebP 3 (correct). Same pattern with loop=3 on m1: GIF 3, APNG 4, WebP 3.

MOT-02 · slides.md quickstart: `\`vixl shape --shape rectangle ...\`` fails (unrecognized arguments: --shape)

MEDIUM · slides/docs

Steps: Run docs/slides.md line 21: `vixl shape --shape rectangle --name band --width 1920 --height 24 --x 0 --y 1056 --fill '#1d3557'`

Expected: The band is added to the master.

Actual: ERROR: unrecognized arguments: --shape. The shape kind is positional (`\`vixl shape rectangle ...\``). A script that keeps going leaves the band out of the deck without saying so.

MOT-03 · Text defaults to white, so the slides quickstart's text is invisible on its #f6f4ef master

MEDIUM · text/slides

Steps: `vixl new 400x200 --background '#ffffff'; vixl text add 'Hi' --name t --size 40; inspect t`. A text op in apply without color does the same. Also run the slides.md quickstart.

Expected: A default text colour that reads on the canvas or page background, or docs that pass --color.

Actual: color is 'white' on white or near-white. The deck contact sheet looks blank, and check reports 1.10:1 contrast errors on every title, body and number.

MOT-04 · ``${page}/${pages}` count hidden pages, so the 3-page PDF reads 1/4, 2/4, 4/4

MEDIUM · slides/pdf

Steps: Deck with pages cover, results, appendix (--hidden) and wrap. Master text `'${page}/${pages}'`. `vixl export deck.pdf; read it with pypdf extract_text`.

Expected: PDF and HTML leave hidden pages out, so numbering should be 1/3, 2/3, 3/3, or the docs should say numbering counts hidden pages.

Actual: PDF pages read '1 / 4', '2 / 4', '4 / 4'. The PPTX keeps the hidden slide and reads 1/4 to 4/4 consistently.

MOT-05 · `\`vixl check --checks motion|character|captions\`` is rejected by the CLI

LOW · checks/CLI

Steps: `vixl -p m2.vixl check --checks motion`

Expected: animation-authoring.md says to run `vixl_check` with the motion, character, anatomy and captions checks, and checks.OPTIONAL_CHECKS includes motion, character and captions.

Actual: usage_error: invalid choice 'motion'. The argparse choices in cli.py leave out motion, character and captions. Python `Project.check(checks=['motion'])` works, and the motion check runs implicitly in a default check.

MOT-07 · line-boil and wiggle default to 1000 ms even in a longer seamless timeline, and line-boil does not fit its variant cycle to the loop

LOW · motion/recipes

Steps: m2.json: 2000 ms seamless timeline with motion line-boil (fps 10) and wiggle, neither given a duration. showcase: line-boil at fps 7 with duration 2400.

Expected: Default to the rest of the timeline, as spin and attach do. In a seamless timeline, choose a hold cycle that closes.

Actual: line-boil keys stop at 900 ms, so the boil freezes for the second half; wiggle keys end at 1000 ms. With duration 2400 at fps 7 and 3 variants, the export warns 'Loop seam: visible on lyr_... ends at False but starts at True'.

MOT-10 · Stagger + repeat in a seamless timeline lengthens the timeline and breaks other loops

LOW · motion/loops

Steps: First version of the showcase: 2400 ms seamless timeline; animate-preset float on d1..d3 with duration 800, repeat 3, stagger 120, close true; text-animate wave 800 ms with repeat

Expected: Staggered repeats wrap around the loop, or the op is refused with a hint.

Actual: Warnings: d2/d3 'cannot be closed', and the timeline grows 2400 -> 2520 -> 2640 ms. Then 'the wave on title repeats every 800 ms, which does not divide the 2640 ms loop'. Workaround: fewer repeats.

MOT-15 · A timeline whose frames are all identical exports as a static WebP with duration 0

LOW · export

Steps: Document with an empty or static 1000 ms timeline; export-timeline x.webp

Expected: duration 1000, matching GIF, which keeps one 1000 ms frame.

Actual: Result: frames 1, duration 0, frame_durations [0]. The file is a non-animated WebP.

MOT-16 · The bounce recipe's keys never touch the ground at contact

LOW · motion

Steps: motion bounce: amount 150, gravity 980, restitution .65, at 20 fps

Expected: A key at or snapped to the contact frame, so the ball visibly lands.

Actual: Keys are sampled every 50 ms, and the closest one is -1.77 px at 550 ms; the second contact is similar. This is minor, but the docs say to inspect the contact frame.

MOT-17 · loop-seam-speed false positive on baked attach tracks of a pendulum

LOW · checks/motion

Steps: m2: arm rotation -30..30 ease-in-out with close; attach sword to the arm tip; Python check(['motion'])

Expected: No speed note, since the arm's speed at the seam is about 0.

Actual: info: 'translate-x on lyr_44a0... speed changes from 12.169 to -12.169 units/s'. This comes from per-frame sampling at the turnaround.

MOT-06 · Loop-seam findings name tracks by layer ID; the linear-motion finding names no layer

UX · checks/motion

Steps: m1/m2/hero: vixl check, or any GIF export of a looping timeline

Expected: Layer names, as the text-animation and poster findings use ('title').

Actual: 'Loop seam: translate-y on lyr_7f9067ccf4ed415f ends at -5.5997...' and 'start on lyr_5541fedd...', with long unrounded floats.

The hero check prints about 15 such lines with only IDs. 'Two-key linear movement starts and stops instantly' names no layer; the source was the spin recipe, which is linear by design.

MOT-08 · Size-target warnings contradict the settings in use

UX · export/size

Steps: big.vixl (1200x800, grain): export-timeline --preset web; --preset chat (showcase); --target-bytes 300000 .webp

Expected: No 'too big' warning when the file meets the preset target, no advice to use a setting already in use, and one warning per problem.

Actual: web: 1,879,995 B fits its 2,000,000 B target but still warns 'GIF is 1.8 MB; ad networks ... cap near 150 KB-1 MB ... pass target_bytes, or WebP measured no smaller here (3,463,542 bytes); export MP4.' The sentence is garbled. The showcase chat GIF reports dither 'ordered' and also warns 'pass dither: ordered'. A missed target_bytes produces two near-duplicate warnings.

MOT-11 · The walk cycle on the default front-facing character reads as leg splits, and no character/anatomy finding is raised

UX · character

Steps: hero.json: character hero, height 320; character-cycle walk 2000/1000; react at 2000; timeline-sheet

Expected: A walk that reads as walking, or a warning that a front view needs a side rig.

Actual: The upper legs rotate +-30 deg in the picture plane while the body stays fixed. At 0.86 s and 1.71 s the feet are about 150 px apart and the arms disappear behind the torso; react splays the feet sideways. Python check(['motion','character','captions']) returns only loop-seam notes.

MOT-13 · The CLI has no WAV export, and Python export_audio refuses to overwrite with a raw OSError

UX · audio

Steps: vixl -p hero.vixl export hero.wav; Python vixl.audio.export_audio(p, existing.wav)

Expected: A CLI route to `vixl_export_audio` or an error that names it, and a structured output `_exists VixlError` as other exports give.

Actual: The CLI says 'Specify a supported export format'. Python raises `FileExistsError [Errno 17]` instead of the structured output `_exists` error.

MOT-14 · `\`drawing smooth\`` after `\`straighten\`` turns square corners into large curves, and the check still passes

UX · drawing

Steps: `sketch.png (house) -> drawing`

`import/clean/vectorize/straighten/smooth/fill; render`

Expected: Smoothing keeps deliberate corners, or the drawing check flags the change in shape.

Actual: The window rectangle becomes a blob, the bottom-right wall corner becomes a curve about 60 px across, and the door top is rounded. `check --checks drawing` only warns 'keeps 89% of the original lines'.

MOT-09 · `animate-preset \`amount\`` means something different for each preset, and this is not documented

DOCS · animate-preset

Steps: `animate-preset core pulse amount 0.08` (meant as +8%)

Expected: Docs and schema describe amount for each preset (pulse/zoom: peak scale; spin: turns; float/shake/bounce: px; blink: count).

Actual: The docs only say 'amount and distance tune them', and the schema field is a bare number. pulse scaled the layer to 0.08 (keys 1.0 -> 0.08 -> 1.0), and the sun core almost vanished.

MOT-12 · Template docs say `\`${title}\`` works but don't say to define the variables first; no warning for an unfitted lyric box

DOCS · lyric video

Steps: Apply a template text `\`${title} / ${artist}\`` in a new document

Expected: The docs mention `\`variable set title ...\``, or the template accepts undefined lyric variables.

Actual: `missing_variable: Undefined variable: title`, and the whole template batch is rejected. After the variables are defined it works. `lyric-video-plan` also returns warnings: [] even though line 1 is wider than the 640 px canvas and is clipped in the video.

Notable passes

- **PASS MOT-18** All six timeline formats have the right frames and timing: GIF 40x50 ms loop 0; WebP 40x50 ms loop 0 (ANIM parse); APNG 40x50 ms; MP4 h264 20/1, 40 frames, 2.000 s; WebM vp9 40 frames, 2.000 s; ZIP 40 PNG + timing.json.

- **PASS MOT-19** chat/web/email presets and `target_bytes` hit their size targets: chat 480x320 15 fps 64 colors 937,620 B; web 800x533 10 fps 1,879,995 B; email 600x400 5 fps 952,710 B (all fits:true). WebP target 300k/60k/5k: the smallest attempt is written (504x336, q45, 10 fps, 482,954 B, 9 tries) with fits...
- **PASS MOT-20** text-animate: all 9 presets with their units, directions and modes: The timeline lists unit counts (e.g. 22 chars, 5 words), stagger and span. Typewriter without stagger spreads 28 units over 500 ms. in-out and the repeating wave give no seam warning, and the others do. The sheet shows the expect...
- **PASS MOT-21** steps(4), cubic-bezier, close, spin symmetry, attach geometry and keyframes sample: stepper x = 0,0,100,100,200,300 (exact). The bezier starts fast (168 px at 240 ms). Attach tip at (203.9, 98), as expected for a 120 px arm at -30 deg. spin symmetry 5 gives 72 deg. The sin sample writes 21 seamless keys.
- **PASS MOT-22** The pixel sprite workflow in `pixel-animation-spacing.md` works as written: `coin.gif 128x128, 3 frames, 350 ms (150/100/100), 4 colours (no smoothing), blink: 4 frames, 520 ms, durations 200/60. loop 2 -> GIF 2 / APNG 3 / WebP 3. MP4 at 50 fps (GCD tick 20 ms), 26 frames, 0.52 s. The sheet JSON includes...`
- **PASS MOT-23** Synth score and SFX are mixed into MP4/WebM/WAV at the requested rates: MP4: h264 72 frames at 24 fps + aac 48 kHz stereo, 3.000 s. WebM reports 24 kHz (the nearest Opus rate above 22050) + opus. 8 kHz AAC is honoured. WAVs are exactly 3.0 s at each rate; 7000 is rejected with a clear range message....
- **PASS MOT-24** lyric-video plan and export with sections, cue, repeat and break: `song.mp4: 108 frames at 12 fps, 9.000 s, aac 44.1 kHz mono. The repeated line is flagged repeat:true. Frames show the intro, next-line preview, chorus background, cue star, the break and 'Goodnight'. Build plus render took 3.3 s...`
- **PASS MOT-25** The fillable PDF has correct AcroForm fields, and `--set` and CSV fills validate: 10 fields: Tx, Btn (radio, Ff 49152), Btn, Ch (Opt with export values and the 'Beef stew' label) and Sig. MaxLen 80/400/5, comb flag, TU tooltips, DR font /VxE1 embedded (DejaVu Sans). The `--set fill` produces a flattened PDF that...
- **PASS MOT-26** Pages, masters, notes, a hidden page and transitions across PDF/PPTX/HTML: PDF: 3 pages, 960x540 pt. PPTX: 4 slides, slide3 show=0, `<p:zoom>/<p:fade>/<p:push dir=u>`, 4 notesSlides. HTML: 3 slides, notes 2. check deck flags the missing notes on the cover. See MOT-02/03/04 for the issues.
- **PASS MOT-27** Sketch import -> clean -> vectorize -> straighten -> fill: 6 traced strokes. The report lists regions with points. A fill point on a line gets a helpful error, and fills at the reported points colour the sun, roof and wall correctly.
- **PASS MOT-28** Text flows across 2 columns and a second page, with overflow reporting: Frames hold 1217/1240/546/581 chars, then 1470/1388 after resizing. `remaining_chars` matches between the operation and the check. The PDF has selectable text over 2 pages.

- **PASS MOT-29** link / stale / refresh / missing / relink / cycle / embed: ok -> stale (check warning) -> link-refresh -> ok. The --set override renders 'LOGO v9'. missing names the path it looked for, and render/export fail with link_missing. relink restores ok. The cycle is refused and the chain named...
- **PASS MOT-30** export_character -> character-load in another document: 20 parts exported (1.9 KB). Loaded with new names and colours, and the run cycle applies.

MCP, REST and security

MCP over stdio and HTTP in every toolset, the REST server, path and SSRF probes, the updater and providers. *14 issues, 12 notable passes.*

SVC-01 · vixl_compose silently makes the composed document active; later calls without document= hit it

LOW · MCP / compose

Steps: Session: create jazz.vixl, then vixl_compose(path='card.vixl', ...), then vixl_export_file(path='out/jazz-job.tiff', as_job=true) and vixl_operations_apply(... target 'title') without document=.

Expected: Either compose leaves the active document alone, or its docs say it switches it (as vixl_document_create does).

Actual: The export wrote card.vixl into out/jazz-job.tiff and the apply ran against card.vixl. Results do name 'document': 'card.vixl', but the compose docs (interfaces.md 'Build a piece in one call') do not mention the switch.

SVC-02 · Undecodable image bytes leak a Python object repr and memory address

LOW · MCP + REST / errors

Steps: MCP vixl_import_image(data_base64=<random bytes>); REST POST /assets?name=x with body 'hello world'.

Expected: A clean message such as 'Cannot decode image: not a supported image format'.

Actual: {"error": "invalid_image", "message": "Cannot decode image: cannot identify image file <_io.BytesIO object at 0x7fc2d83485e0>"}

SVC-03 · Compact toolset instructions and tool descriptions name tools it does not serve

LOW · MCP / compact toolset

Steps: Start vixl mcp --tools compact; compare vixl_* names in initialize.instructions and tool schemas with tools/list.

Expected: Only served tools are named (CLAUDE.md: tool descriptions must not mention unknown vixl_* names).

Actual: Instructions name 11 unserved tools (vixl_check, vixl_sizes_list, vixl_layouts_list, vixl_fonts, vixl_font_pair, vixl_styles, vixl_capabilities, vixl_roll, vixl_brushes_list, vixl_timeline_preview, vixl_export_timeline). The as_job field descriptions of vixl_operations_apply/import_document/workflow say 'poll vixl_job', which compact lacks. The runtime job pointers do correctly say vixl_workflow(action='status'), and that works. core and ai modes are clean.

SVC-08 · MCP initialize reports serverInfo.version 1.30.0 (the MCP SDK version), not Vixl 0.22.0

LOW · MCP / serverInfo

Steps: Initialize any vixl mcp server and read serverInfo.version.

Expected: 0.22.0, so clients and logs show the engine version.

Actual: "version": "1.30.0" in core, compact and ai modes.

SVC-13 · 401 responses carry no WWW-Authenticate header

LOW · REST / auth

Steps: VIXL_API_TOKEN=... vixl serve --host 0.0.0.0; GET /document with no token or a wrong token.

Expected: 401 with 'WWW-Authenticate: Bearer' (RFC 6750).

Actual: 401 {"error": "unauthorized"} with no header. The token itself is enforced correctly (docs, openapi and APIs are protected, /view is public, ?token= is not accepted).

SVC-14 · Oversized request bodies are rejected only after being read

LOW · REST / limits

Steps: POST /operations with a 200 MB JSON body; POST /assets with 60 MB of zeros.

Expected: An early 413 based on Content-Length.

Actual: 200 MB: 413 {"error": "resource_limit"} after 3.2 s, and server RSS was about 345 MB afterwards, so the body appears to be buffered first. 60 MB of zeros was accepted and decoded (invalid_image), with no size error. No crash or hang.

SVC-04 · Tool argument validation errors are raw pydantic text, not the documented JSON error shape

UX · MCP / errors

Steps: vixl_resource_get(name='../../etc/passwd') with the required 'kind' missing.

Expected: JSON {error, message, field, suggestions} like every other failure.

Actual: '1 validation error for vixl_resource_getArguments/nkind/n Field required [type=missing ...] For further information visit <https://errors.pydantic.dev/2.13/v/missing>'

SVC-05 · Did-you-mean for unknown fields ignores aliases ('colr' suggests 'control', not 'color'/'fill')

UX · MCP / errors

Steps: `vixl_operations_apply` with `{type: shape, colr: 'red'}`.

Expected: Suggest 'fill' (the field that 'color' normalizes to).

Actual: "Did you mean 'control' instead of 'colr'?" The multi-error report itself is good: 2 invalid ops listed, nothing applied.

SVC-06 · Several MCP errors point to CLI commands and give empty suggestions

UX · MCP / errors

Steps: `vixl_compose` with layout name 'no-such-layout'; `layout-apply` with an unused slot; `vixl_check` fonts finding.

Expected: MCP-facing text names MCP tools (`vixl_layouts_list`, `vixl_font_pair`) and `unknown_layout` suggests close names.

Actual: 'Unknown layout ...; see vixl layout list', suggestions: []; 'vixl layout show NAME lists each slot's meaning'; 'Choose typefaces: vixl font pairings / font pair, or font install.'

SVC-07 · POST /export rejects unknown fields without naming them; POST /compare silently ignores them

UX · REST / errors

Steps: `POST /export` `{"path": "out/x.png"}` and `{"format": "PNG", "bogus": 1}`; `POST /compare` `{"bogus": "x"}`.

Expected: Name the offending field and list the accepted ones (as `POST /preview` does); consistent unknown-field handling.

Actual: `/export`: 400 `{"error": "invalid_operation", "message": "Unknown export option"}` with no field. `/compare`: 200, field ignored. `/preview`: 400 listing the accepted keys (good).

SVC-11 · Linux pip install: every update command fails with exit 1 and points to a Windows installer

UX · Updater

Steps: `vixl update --check`; `vixl updates status` (`--json` too); `vixl update`; `vixl updates on`; `vixl update --rollback` (Linux `venv` from the release wheel).

Expected: A clean message saying pip installs are updated with pip (e.g. 'pip install -U ...'). 'updates status' and '--check' could still report the installed and latest version.

Actual: All exit 1: 'Automatic updates require the Windows installer. Python/pip installs remain managed by pip; download Vixl-Setup from <https://github.com/jxburros/Vixl/releases/latest>'. No traceback and no network or file side effects, which matches `releases.md` ('explanatory error'). A Linux user is still pointed to a Windows-only installer.

SVC-09 · Streamable HTTP section contradicts the per-session active document

DOCS · Docs / interfaces.md

Steps: Two Streamable HTTP clients on one `vixl mcp --http server`: A creates `http-a.vixl`, B opens `jazz.vixl`, both inspect without `document=`.

Expected: Docs agree with 'Several agents on one server': the active document is per client session.

Actual: Measured per-session (A: `http-a.vixl`, B: `jazz.vixl`), as the 'Several agents' section says, but the 'Streamable HTTP and live review' section says 'A shared server has one active-document selection'.

SVC-10 · Default detail='brief' returns full text metrics, not the documented minimal fields

DOCS · Docs / MCP detail

Steps: `vixl_operations_apply` with a `layout-apply` (8 layers) and default detail.

Expected: Per the docs, `brief` returns new layers as `added/name/type/bounds`.

Actual: Each new text layer also carries `ink_bounds`, `line_bounds`, `baseline`, `baselines`, `ascent`, `descent`, `cap_height`, `x_height` and `metrics_space`. Not harmful, but heavier than documented.

SVC-12 · 'vixl providers' supported list does not match providers.md

DOCS · Providers

Steps: `vixl providers` with no configuration.

Expected: Same names as `providers.md` (`openai`, `gemini`, `flux`, `anthropic keyless`, plus `automatic1111/comfyui/http gateway` via config).

Actual: supported: [`openai`, `anthropic`, `mistral`, `meta`, `gemini`]. `flux` is missing, and `mistral/meta` are not documented in `providers.md`. 'ai generate --provider local' gives 'Unknown provider: local' without listing the configured or available names. The `provider_not_configured` errors are clean otherwise.

Notable passes

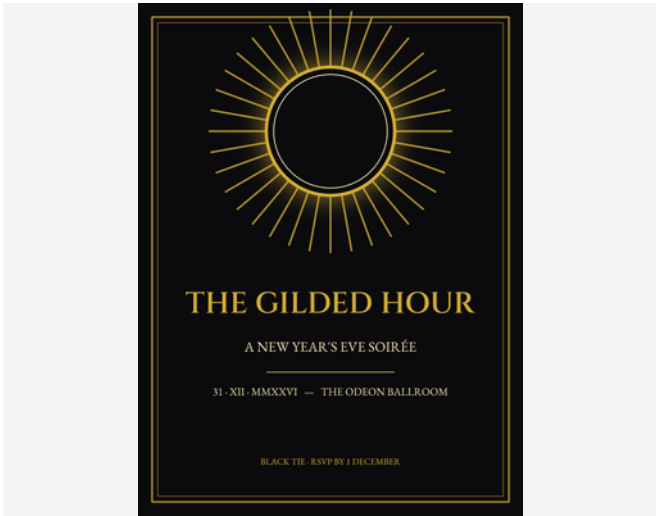
- **PASS SVC-15** tools/list sizes and counts match `CHANGELOG` and `mcp-toolsets.md` exactly: default=core/slim 51 tools 58,141 chars; all/full 62 / 146,671; all/slim 62 / 64,762; core/full 51 / 140,050; compact/slim 13 / 21,653; compact/full 13 / 103,562; ai 16 / 11,045. Startup about 0.9-1.5 s; list_tools 4-52 ms.
- **PASS SVC-16** Realistic stdio session completes with check and preview findings; showcase rendered: All succeeded. The dry run left history unchanged; apply took 0.78 s, check 0.31 s, preview 0.18 s, PNG export 0.15 s, compose with check, preview and export 1.38 s. The useful 'unused_slot' error caught copy that would have been...
- **PASS SVC-17** Error JSON shape and unknown-tool messages: Shape as documented, and the multi-error batch lists every invalid op. `vixl_text_add` names the 'text' operation replacement; `vixl_render` → 'Did you mean `vixl_render_preview...`'; `vixl_ai_generate` → 'served by vixl mcp --tools ai';...

- **PASS SVC-18** request_id retries never apply twice: Replay took 22 ms and returned the recorded result; different args gave request_id_conflict; history showed exactly one revision. A failed call's id is not remembered (as documented). A retry with the same id while the job runs r...
- **PASS SVC-19** as_job and automatic jobs: Auto-job after 1.0 s with progress (1778/4000), completed in 6.3 s; the failed job returns the full error; an unknown id gives not_found. In compact mode the job pointers say vixl_workflow(action='status') and that works.
- **PASS SVC-20** --http Streamable HTTP mode: 51 tools, list in 36 ms; per-session active document; 6 parallel previews OK; bad Host → 400, foreign Origin → 403; '--host 0.0.0.0' → 'ERROR: Non-loopback serving requires VIXL_API_TOKEN'.
- **PASS SVC-21** Workspace confinement over MCP and REST: All 30+ probes were refused ('Path is outside the workspace', with no absolute paths in the errors); no files were created outside. Filesystem fields in operations are refused. vixl_workspace_list hides symlinks.
- **PASS SVC-22** URL imports refuse private targets in every encoding; redirects re-checked: All refused with unsafe_url naming the resolved address; the listener logged 0 connections. Every redirect hop is re-validated, http downgrade is refused, more than 5 redirects fails, and a host with any private A record is refus...
- **PASS SVC-23** Malicious .vixl archives: zip-slip → invalid_project 'Unsafe or unsupported archive member'; linked → linked_asset_disabled at render/export; traversal asset → missing_asset. Nothing was written outside.
- **PASS SVC-24** vixl serve endpoints, bind, auth, CORS: Default bind 127.0.0.1 (docs: 8765). No auth by default on loopback, with Host-header validation (evil Host → 400). Every request with a foreign or null Origin is refused 403 cross_origin_forbidden (also text/plain and form 'simp...
- **PASS SVC-25** Export refuses to overwrite without overwrite=true: 'Destination already exists; set overwrite=true'. Export to a .vixl name is refused.
- **PASS SVC-26** AI commands without providers fail cleanly: provider_not_configured: 'Configure a provider in ~/.config/vixl/providers.json or set VIXL_AI_PROVIDER'; '--provider openai' → 'Set OPENAI_API_KEY for provider openai'. Minor: 'vixl ai' without a project says 'No current project...

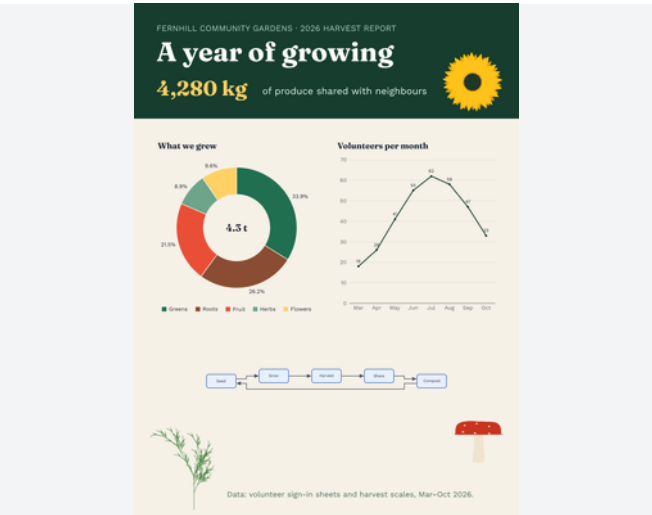
Made with Vixl during testing



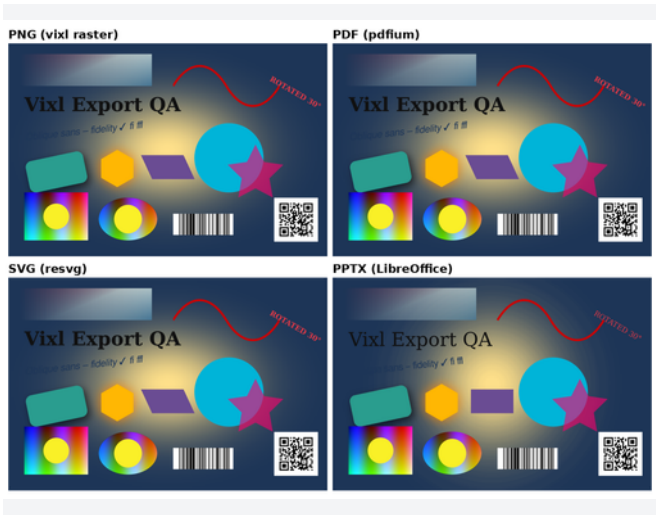
Mandala: radial-repeat rings, gradient petals and a glow core (creative).



The Gilded Hour: art-deco poster with a sunburst and a font pairing (creative).



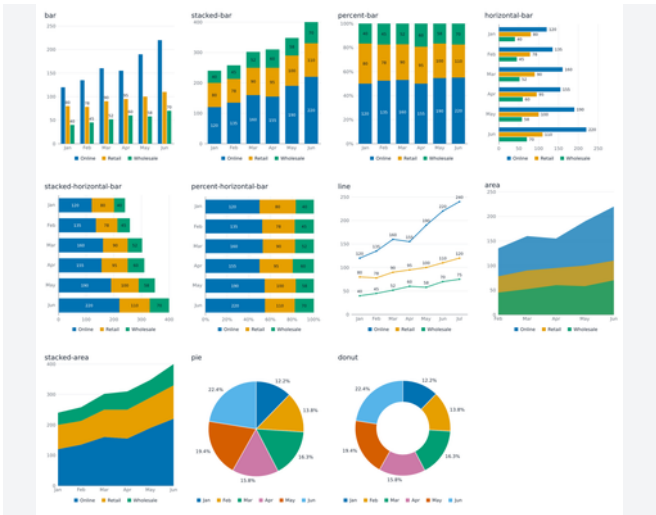
Harvest infographic: donut and line charts, diagram-from-text, organic plants. The small diagram shows CRE-04.



One document, four renderers: PNG, PDF (pdfium), SVG (resvg), PPTX (LibreOffice). The PPTX drops skew (EXP-04) and bold (EXP-05).

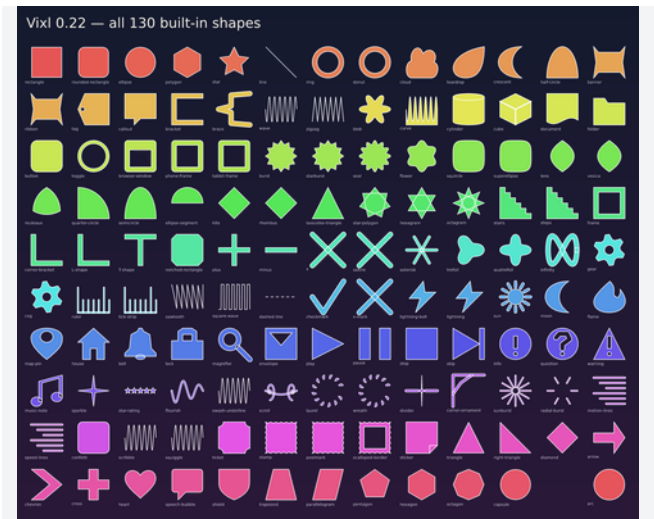


All 33 organic presets, each from a single operation (creative).

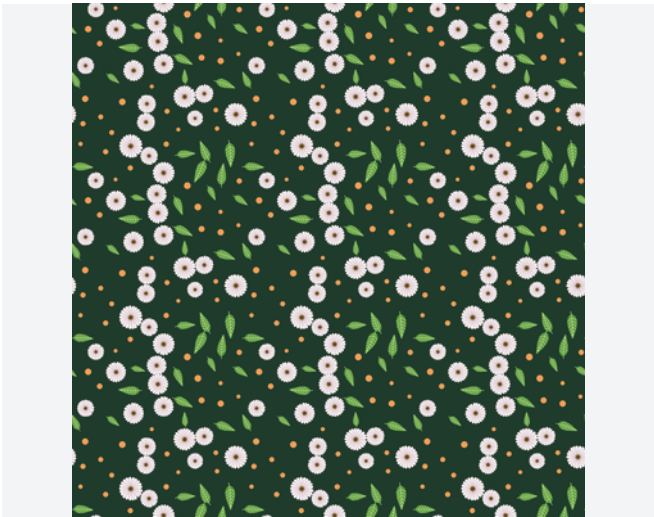


All 11 chart kinds rendered from CSV (creative).

Gallery, continued



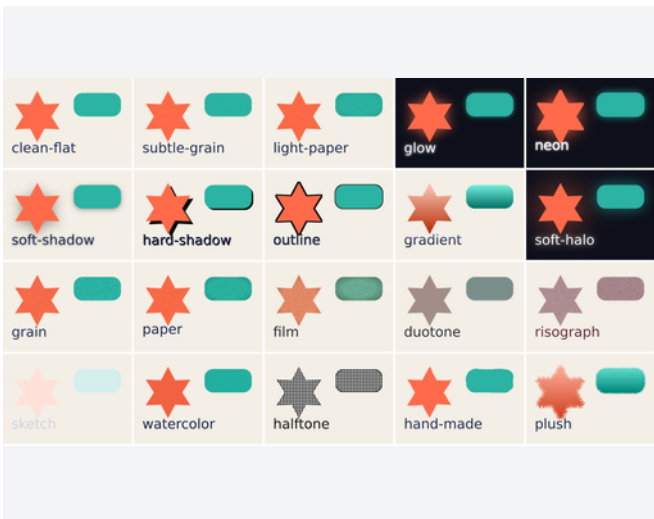
All 130 built-in shapes (core). The confetti shape ignores count (CORE).



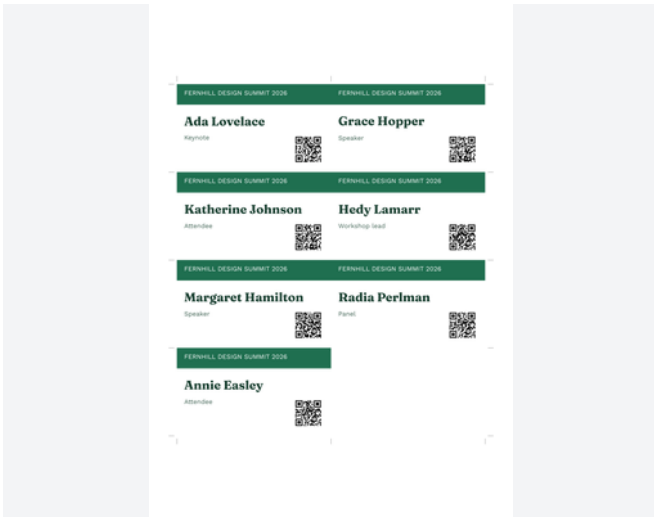
pattern-scatter tile repeated 3x3: no visible seams.



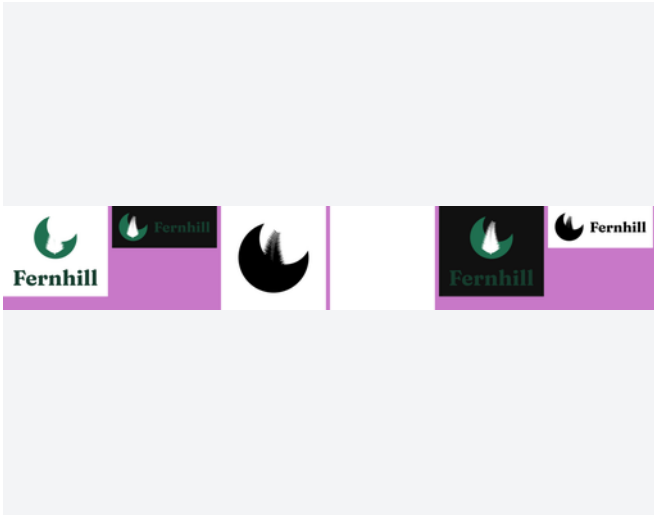
Seamless 2.4 s loop: spinning sun, bouncing ball, line-boil and wave text (showcase.gif in the zip).
Vixl 0.22.0 · QA, stress & exploration report · 7 October 2026



Finishing looks: glow, neon, risograph, halftone, plush and more. Sketch is a faint wash (CRE-07).

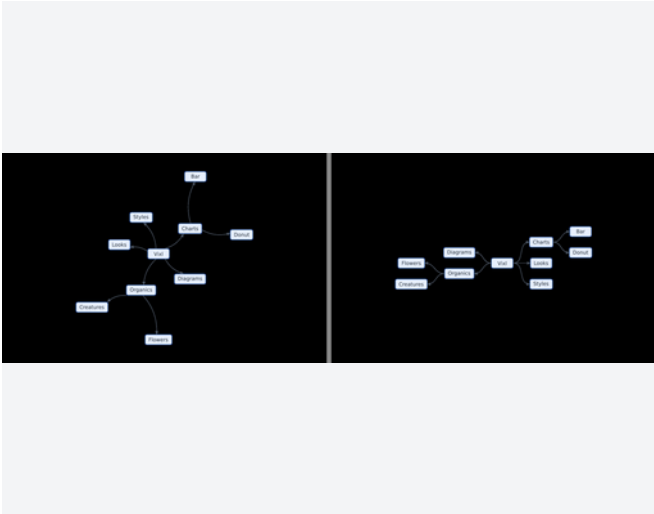


merge-impose: 7 badges from a CSV on A4 with crop marks and a QR code per row.

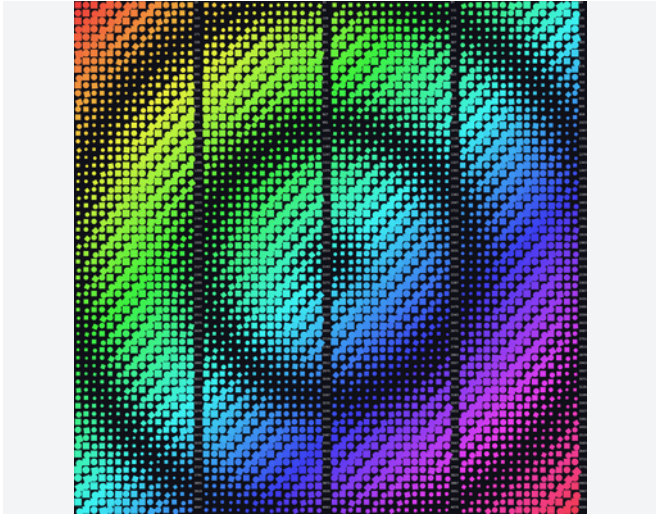


logo-package variants: background baked in and contrast logic inverted (CRE-01/02).

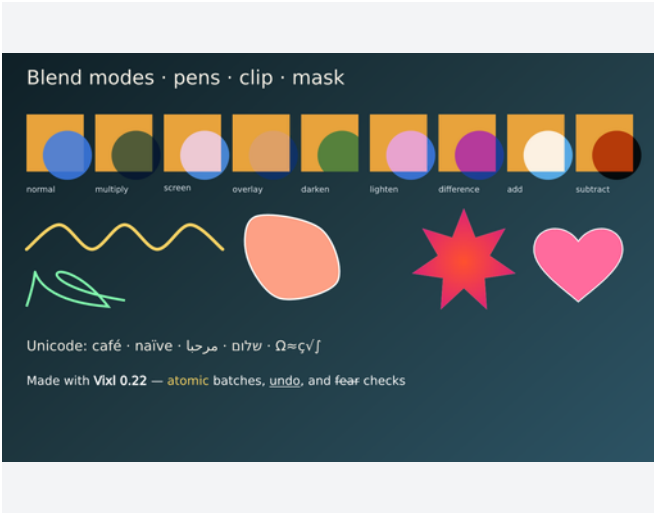
Gallery, continued



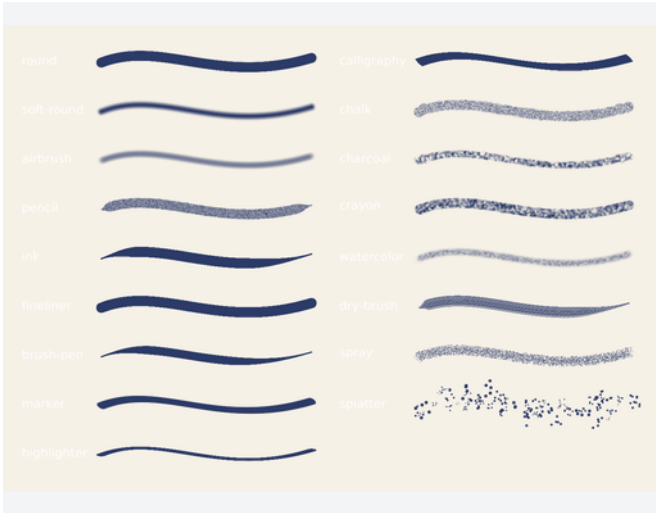
Cyclic diagram layouts: back-edges cross every node (CRE-03).



4,096 layers, the documented maximum: render 11.2 s, peak 228 MB (stress).



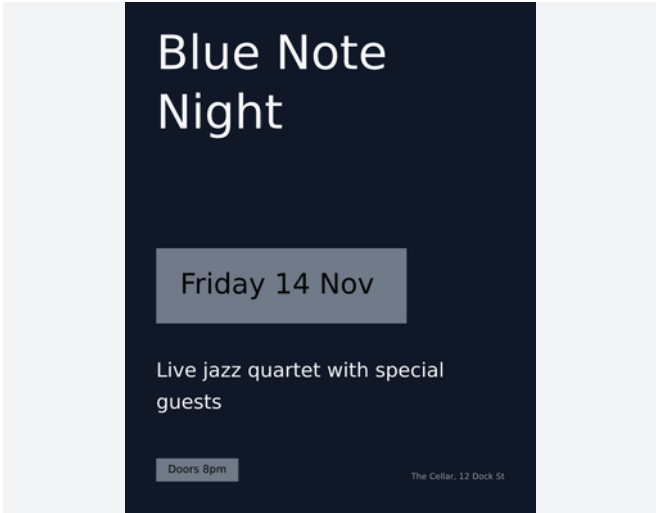
Blend modes, pen paths, clipped gradients, rich and bidirectional text (core).



All 17 brushes; the highlighter is opaque (creative).



Brutalist style: the style check misreads Archivo Black as weight 400 (CRE-05).



Built entirely over MCP: guide, create, apply, check, preview, export (services).

How this report was made

Made with Vixl

This report is a single **.vixl** document built with the release under test through its Python API (`build_report.py`, included in the zip):

- one A4 canvas at 150 dpi;
- a master page with the header strip, footer rule and the page-number variables;
- the IBM Plex Mono and Sans font pairing;
- native chart groups;
- one text-flow story that adds pages until the flow stops overflowing;
- an image grid.

It was exported to PDF, PPTX and HTML, and rendered as a contact sheet.

Building it also turned up findings LEAD-07, LEAD-08 and LEAD-09: fractional widths and literal placeholders. Producing it took about two minutes, mostly in text-flow measurement.

What the zip contains

- **report/**: this PDF, the editable **.vixl** master, the PPTX and HTML versions, the contact sheet and the build script.
- **findings/**: all 177 findings as JSON plus a combined CSV.
- **lead/**: release checks, test-suite log and the open-path repro renders.
- **core/ export/ stress/ creative/ motion/ services/**: every document, render, export, animation, audio file, script and log each area produced, including the minimal repros under **repro/**.

Suggested fix order

1. EXP-01: use `geometry.default_fill` in `vector_strokes.py`. It is a one-line fix and the most visible defect.
2. Validate at apply time anything that later breaks rendering: fractional rich-text width, blur radius, and integer-only sizes generally.
3. Make apply cost independent of unrelated layers: run inspect once and cache text metrics.
4. Fix the logo-package background and contrast handling.
5. Bring SVG, PDF and PPTX into line with the PNG output:

overflow, masked opacity, skew, font style, blend modes.
6. Default text colour, the slides quickstart, and the documentation drift listed under DOCS findings.